

CLIMATE CHANGE

The Heat Is On: Our Planet is Warming

A Story About Our Planets Rising Temperature and Its Impact

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WHAT IS CLIMATE CHANGE?



Temperature

Human Activities

Burning Fossil Fuels

Trap Heat in the
Earth's Atmosphere

Climate change refers to long-term shifts in temperature and weather patterns. Human activities – especially burning fossil fuels – release gases that trap heat in the atmosphere.



ROLE OF GREENHOUSE GASES

Greenhouse Gases: Carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O)

They trap heat in the atmosphere, creating the warming effect we call the greenhouse effect



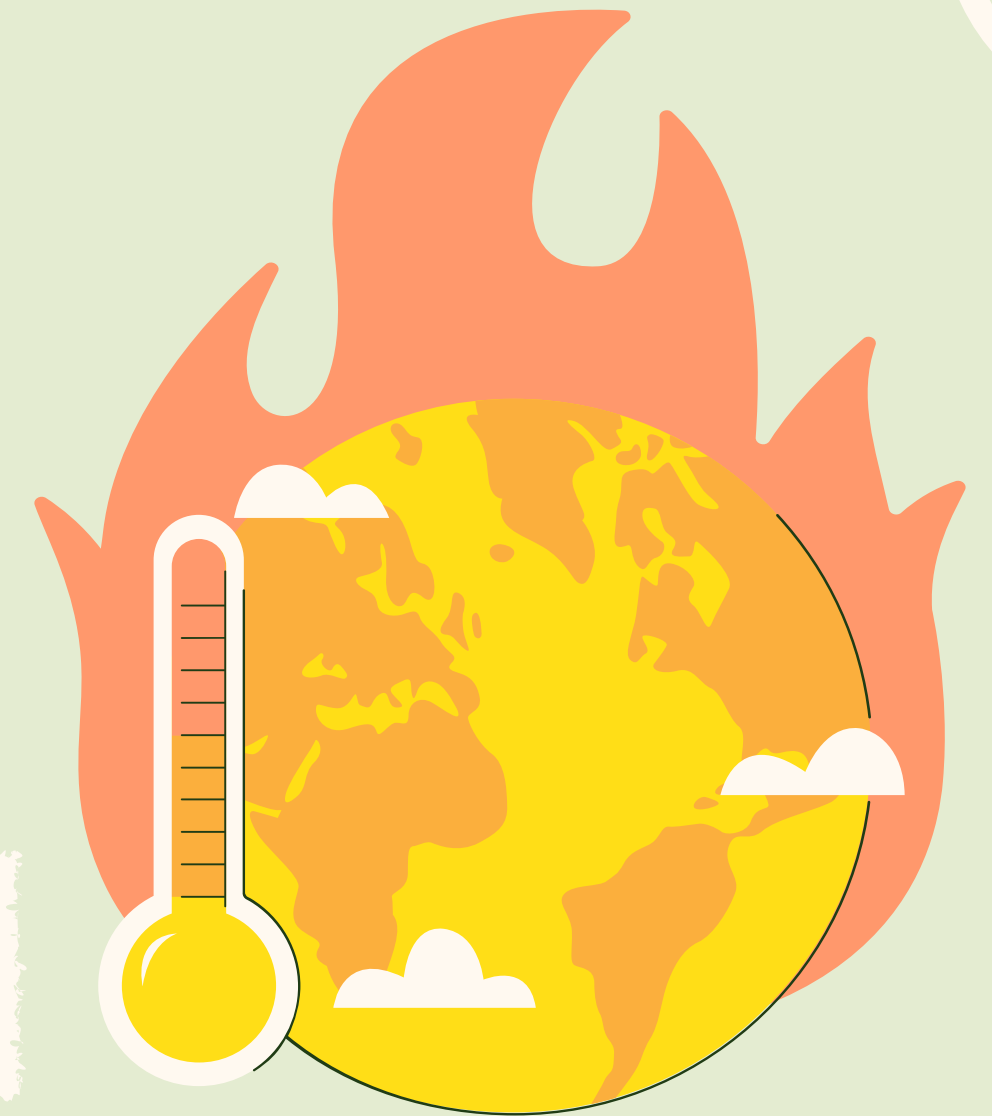
CAUSES OF CLIMATE CHANGE

Deforestation

Agriculture

Burning Fossil Fuels

Industry and
Transportation



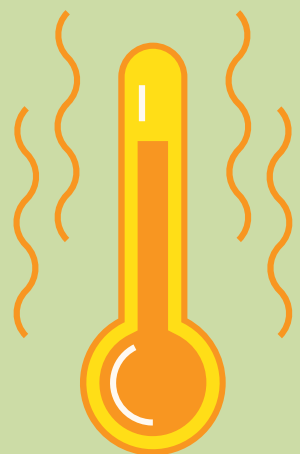
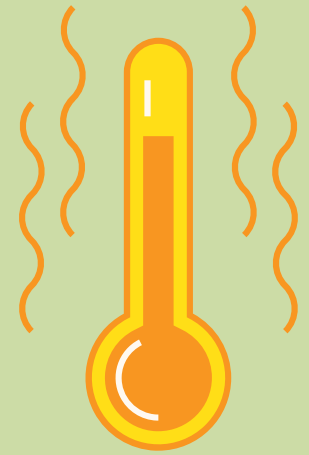
The main causes of climate change come from human activity. Burning fossil fuels for energy and transportation produces large amounts of CO₂. Industries release additional emissions. Deforestation reduces the number of trees that can absorb carbon. And agriculture, particularly livestock, adds methane to the atmosphere.



IMPACT OF GLOBAL WARMING

Increase in Global Temperature: Rising average temperatures on Earth

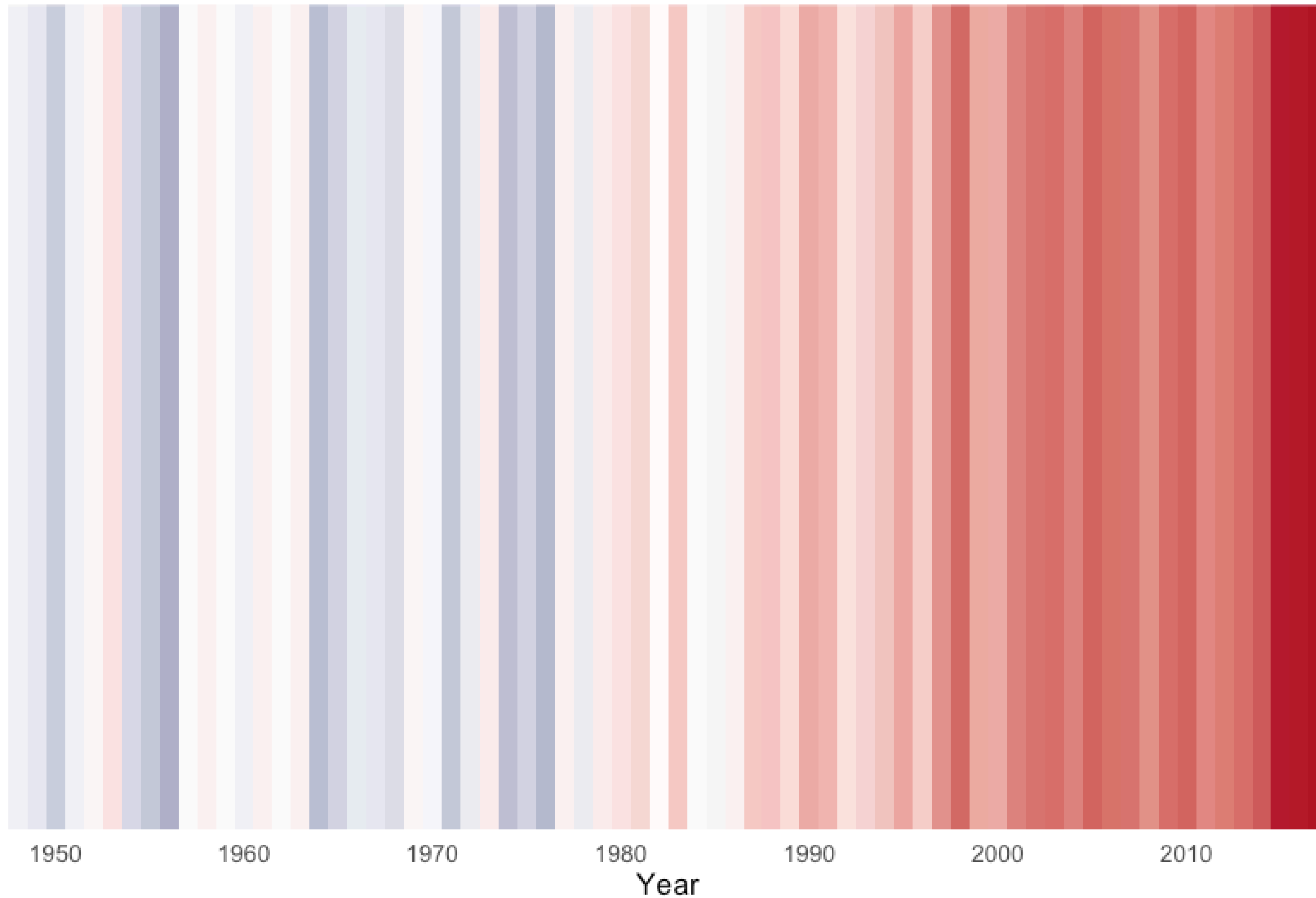
Changing Weather Patterns: More frequent extreme weather events



CLIMATE STRIPES



Global Climate Stripes (1948–2017)
Median global temperature (-0.27°C to 0.79°C)

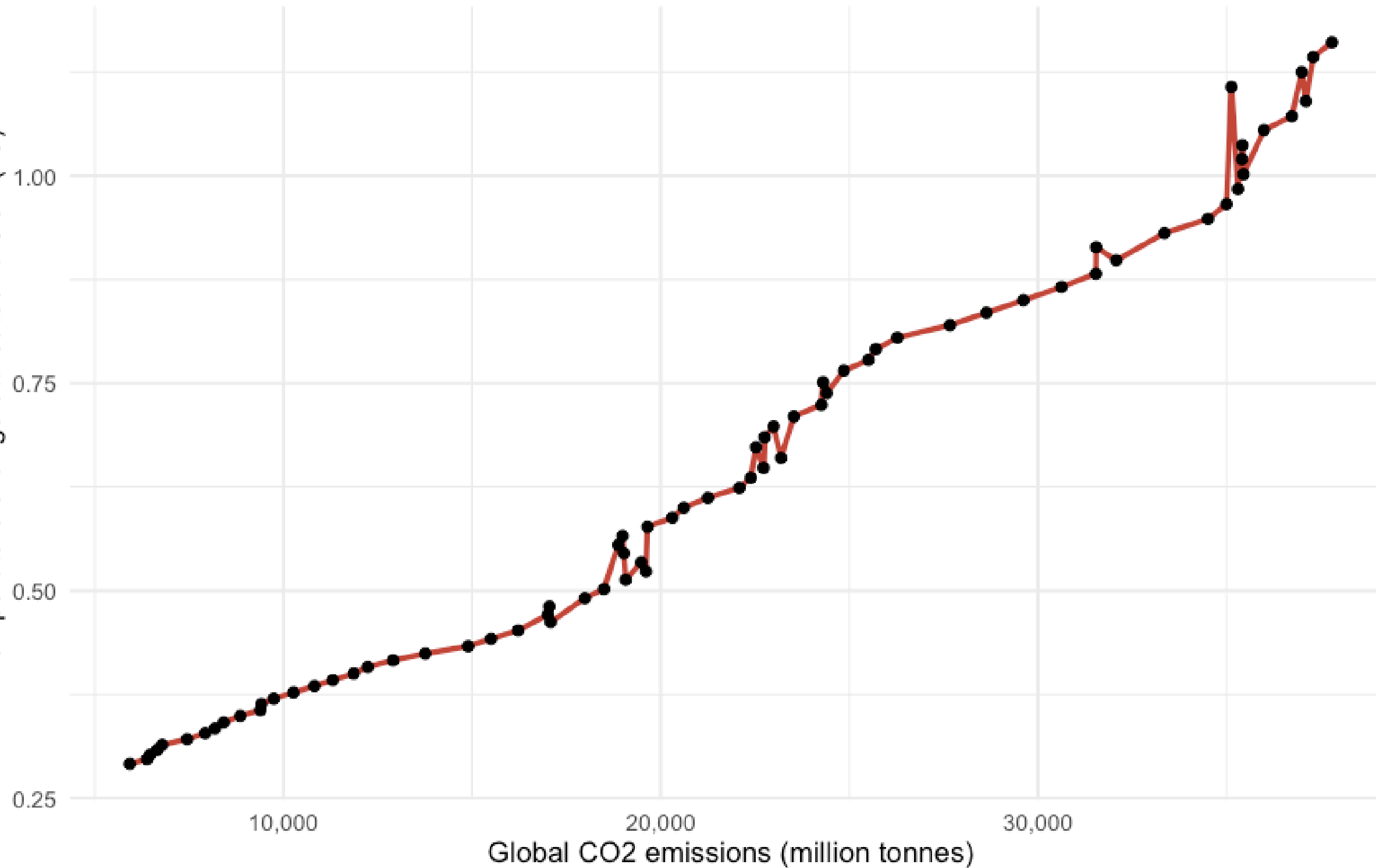


CO₂ EMISSIONS VS TEMPERATURE CHANGE

Global CO₂ Emissions and Contribution to Temperature Change (1950–Present)

World series from OWID; higher emissions align with greater warming contribution

Temperature change attributed to CO₂ (°C)





IMPACT ***MELTING POLAR*** ***ICE & GLACIERS***



Ice Melting: Rising sea levels

Flood Risks in Coastal Areas

Drought and Dry Land



IMPACT *DISRUPTION OF* *ECOSYSTEMS*

Habitat Changes: Many species are at risk

Increased risk of extinction for plants and animals

Forests, oceans, and wetlands are changing rapidly





IMPACT ON HUMANS & HEALTH



Heatwaves increase health risks

Threats to agriculture and food supply

Worsening air quality increases respiratory issues.

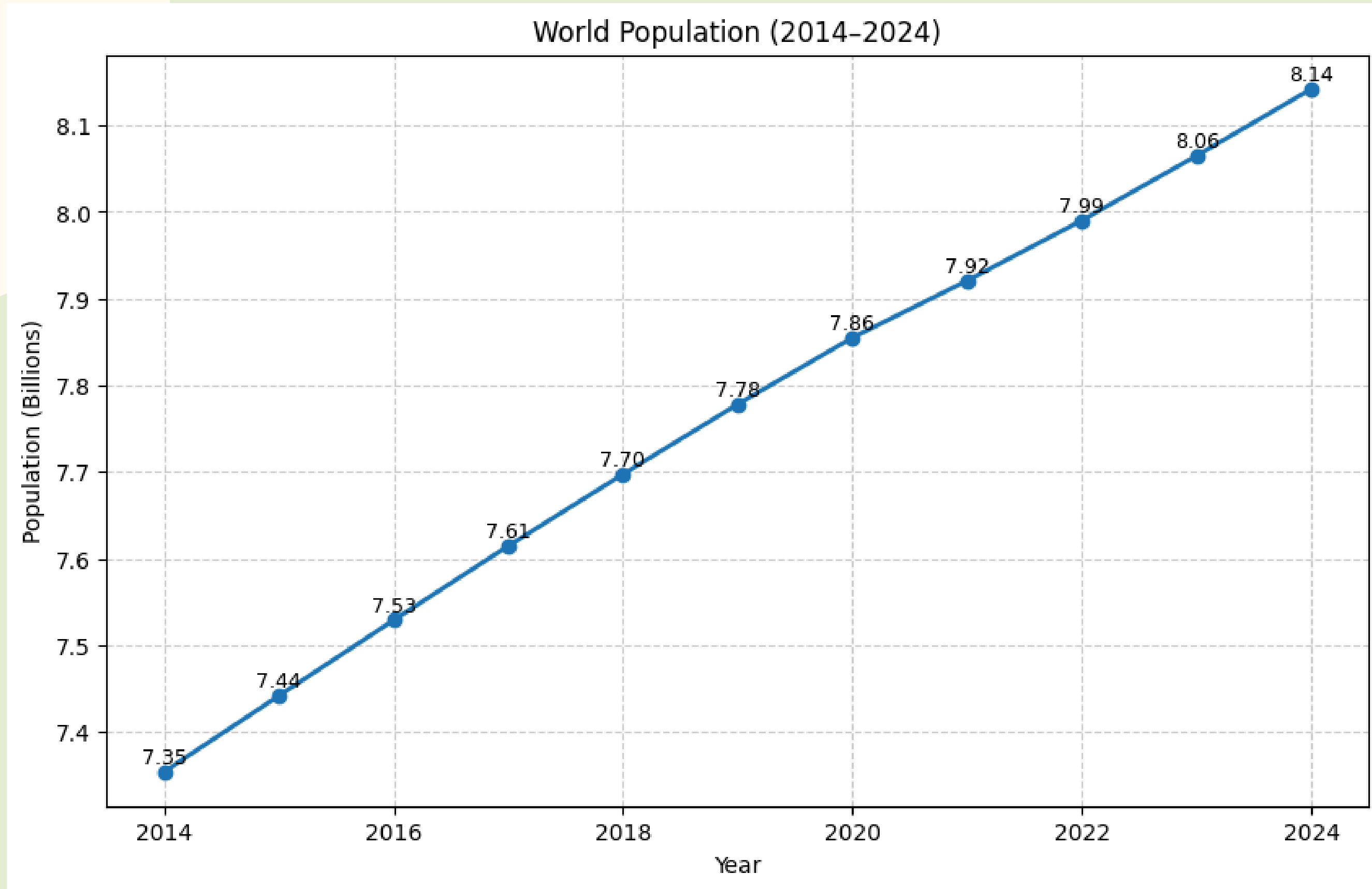


THE EARTH IS WARMING — BUT IS ECONOMIC PROGRESS MAKING IT WORSE?

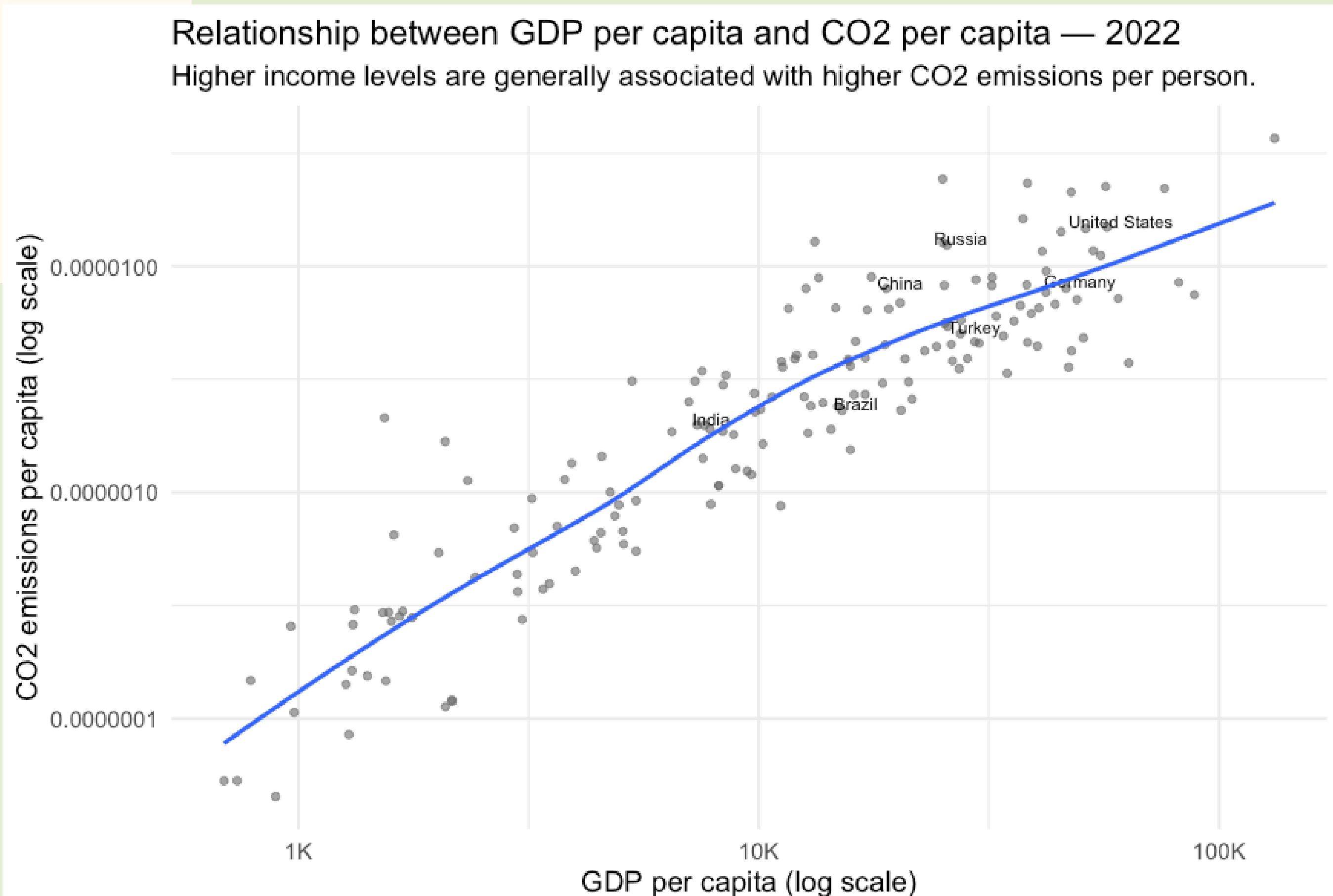
The Earth is clearly warming – but is economic progress making it worse?
To understand this connection, we need to look at population, development,
and emissions.



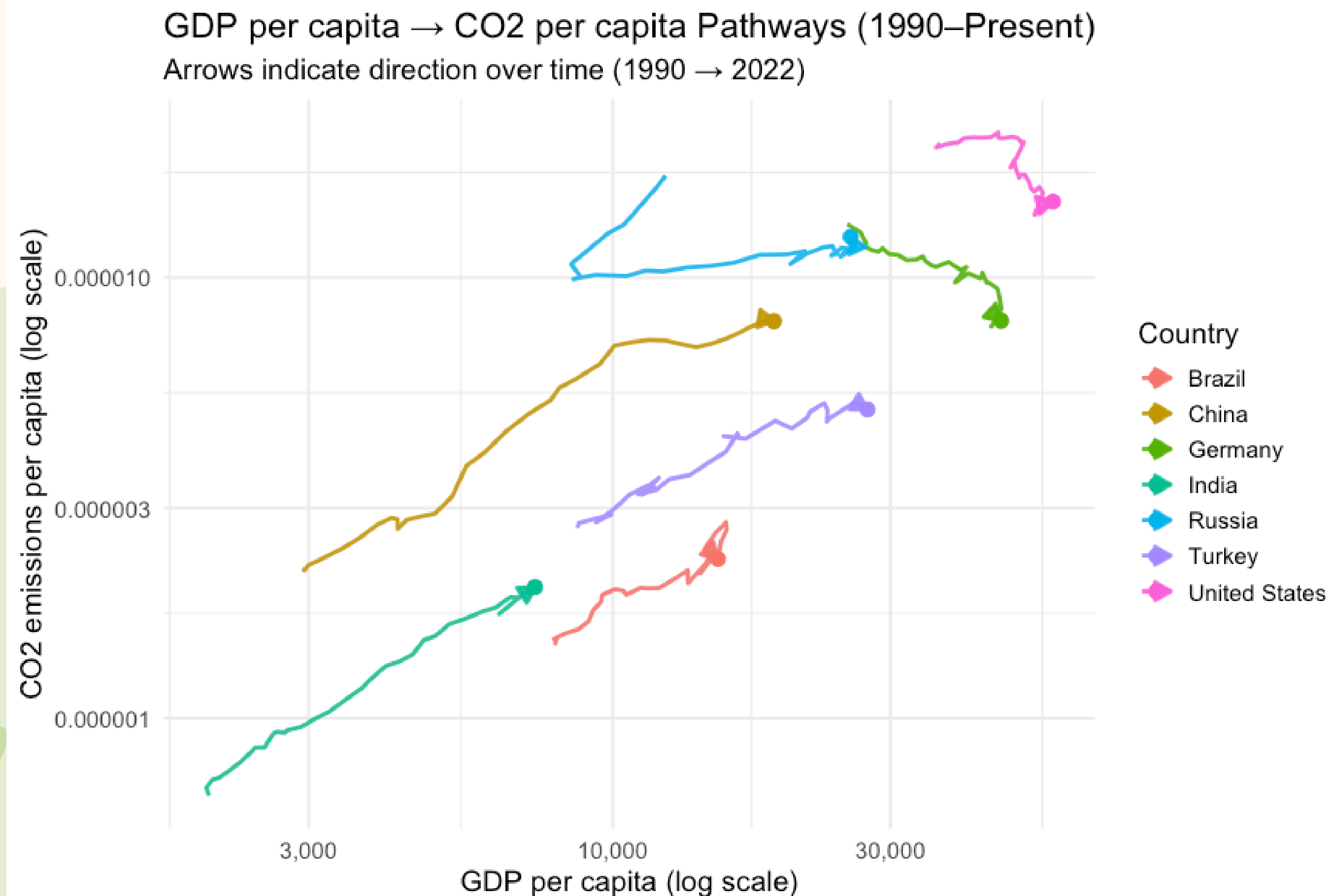
POPULATION GROWTH (2014- 2024)



GDP PER CAPITA VS CO₂ PER CAPITA (2022)

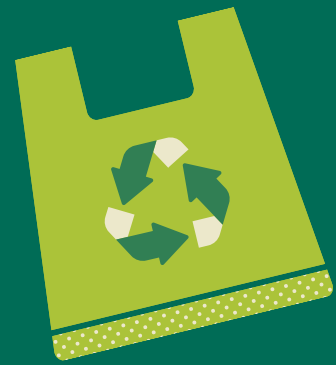


GDP & CO₂ PATHWAYS OVER TIME





BUT....WHO'S RESPONSIBLE?



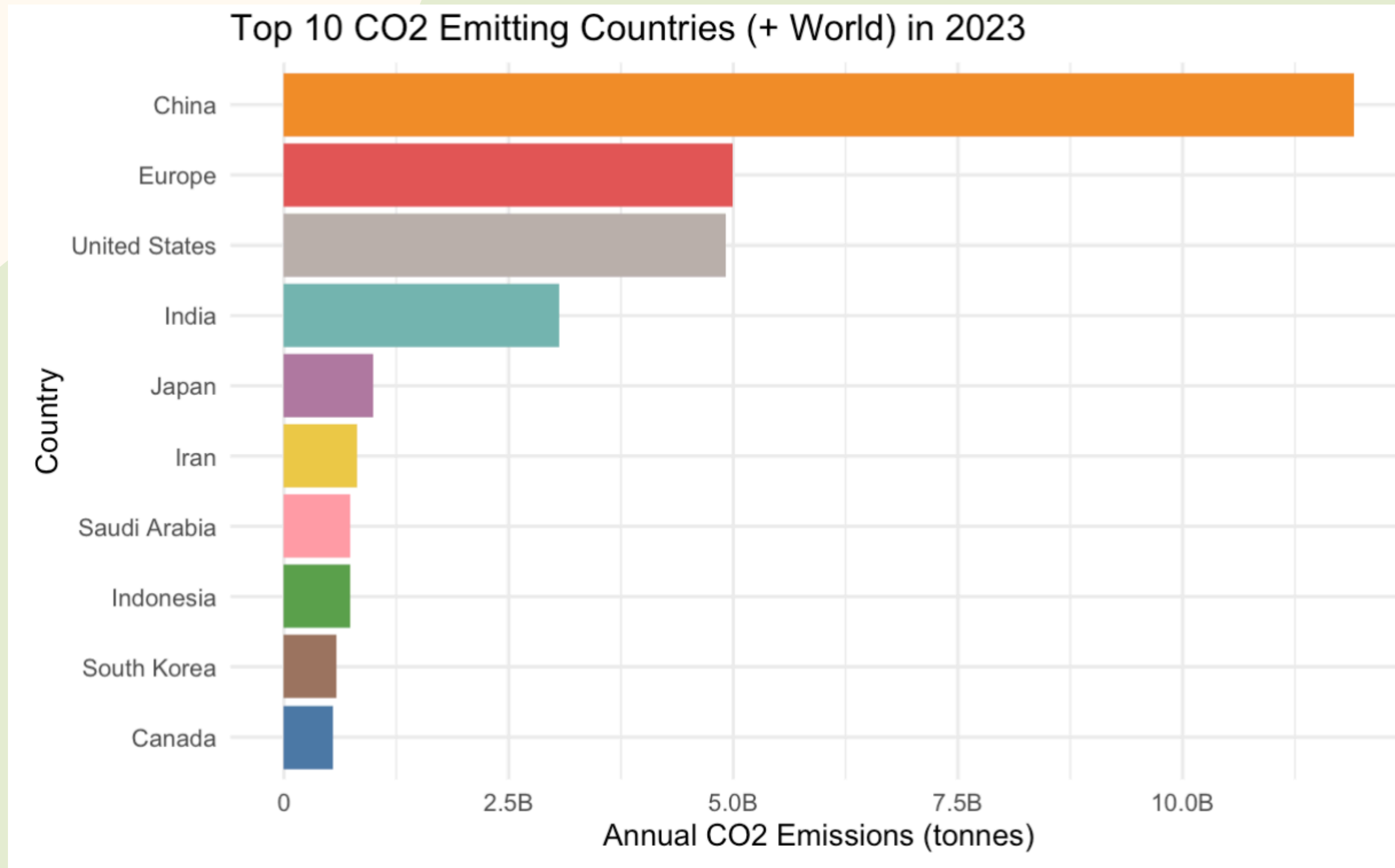
Q1: Which Countries Emit The Most Carbon Dioxide Today?

Q2: Which Countries Have Emitted The Most Carbon Dioxide In Total?

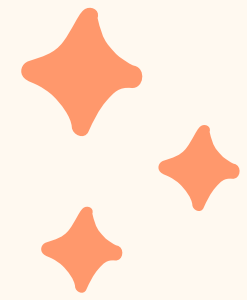
Q3: Which Countries Emit The Most Carbon Dioxide Per Capita?



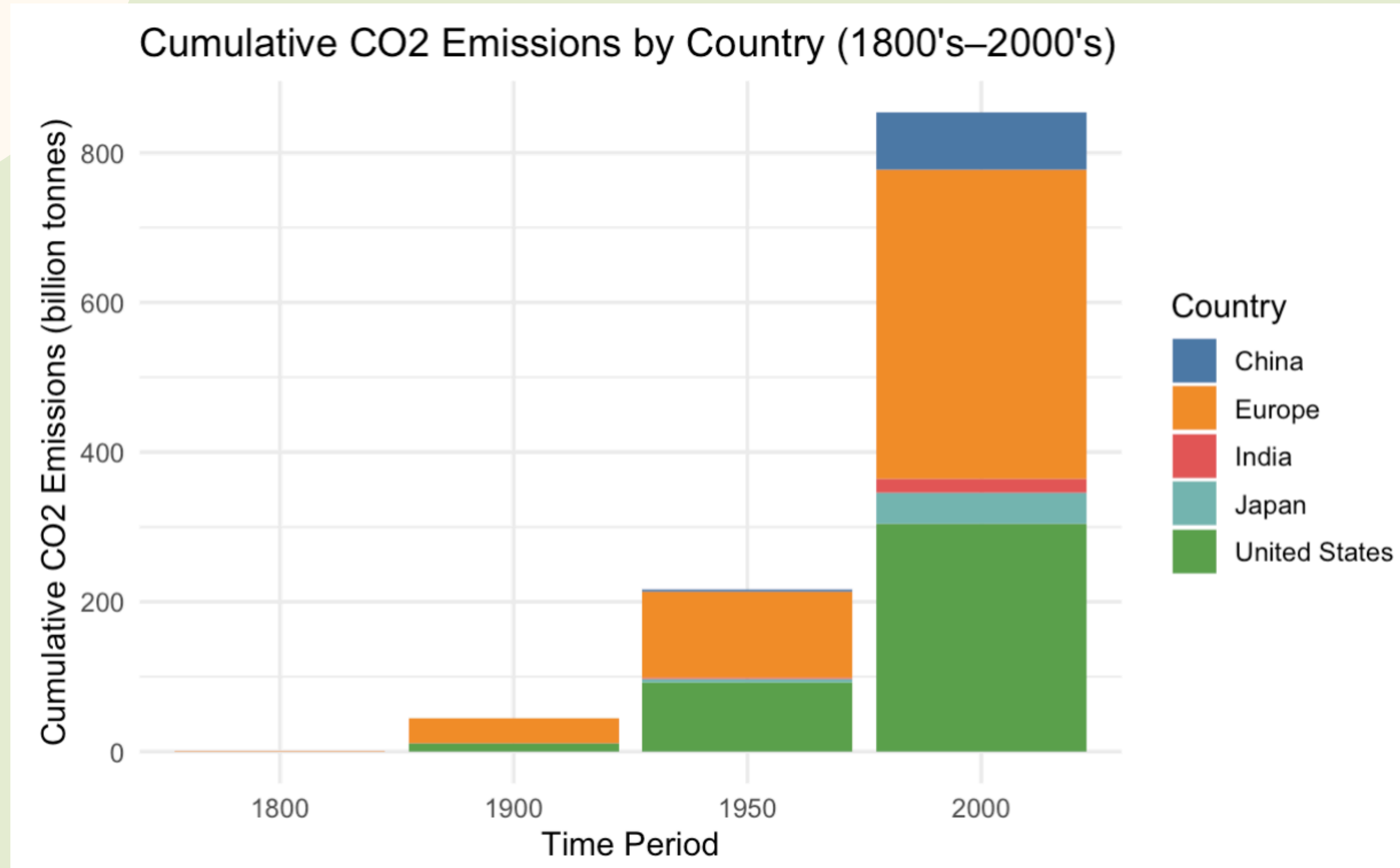
Q1: WHICH COUNTRIES EMIT THE MOST CARBON DIOXIDE TODAY?

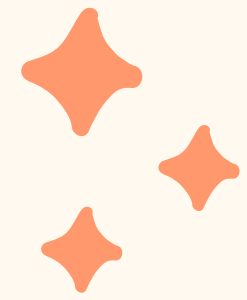


**KEY
TAKEAWAY:**
**CHINA +
EUROPE +
US =
> 60%!**

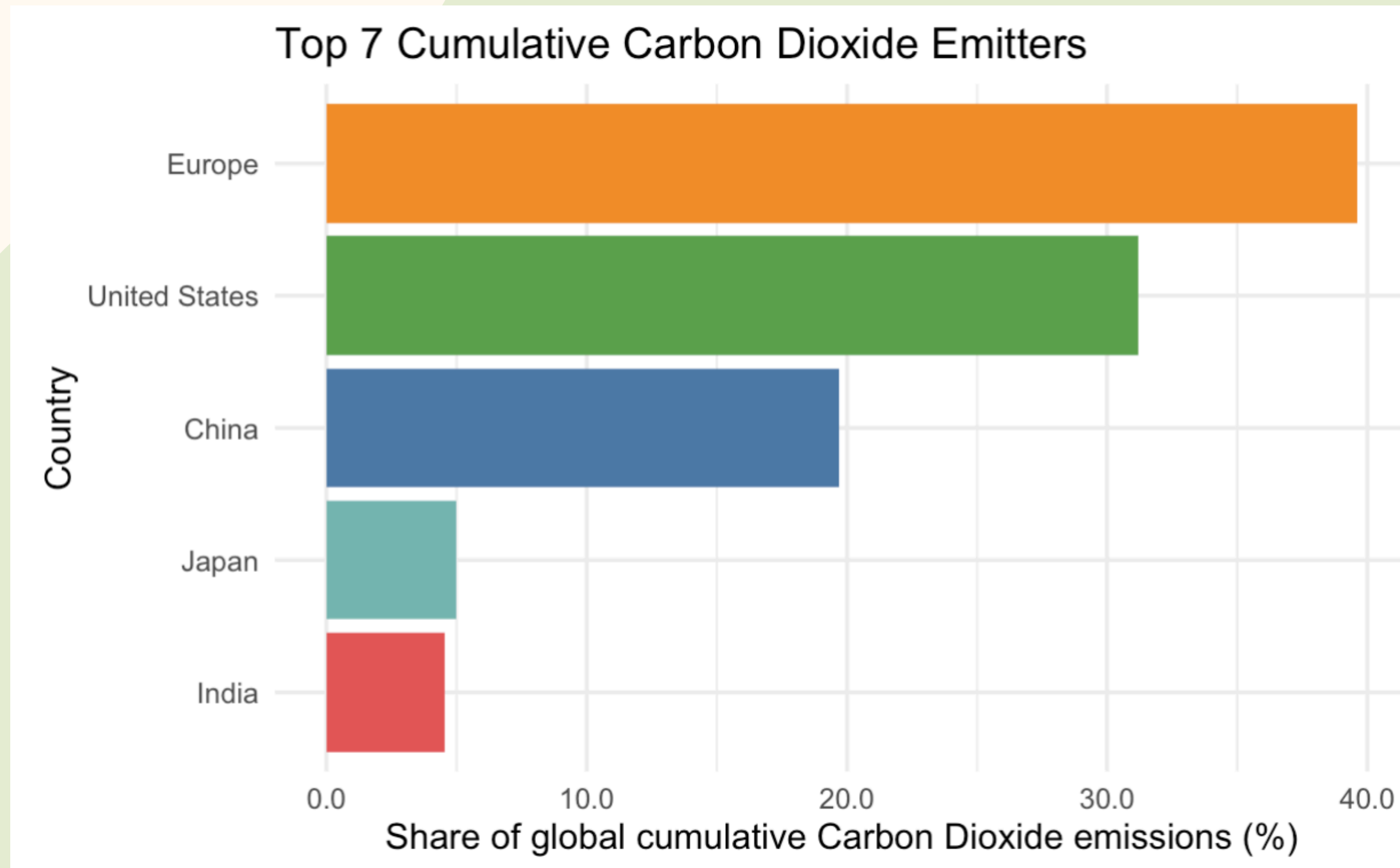


Q2: WHICH COUNTRIES HAVE EMITTED THE MOST CARBON DIOXIDE IN TOTAL?



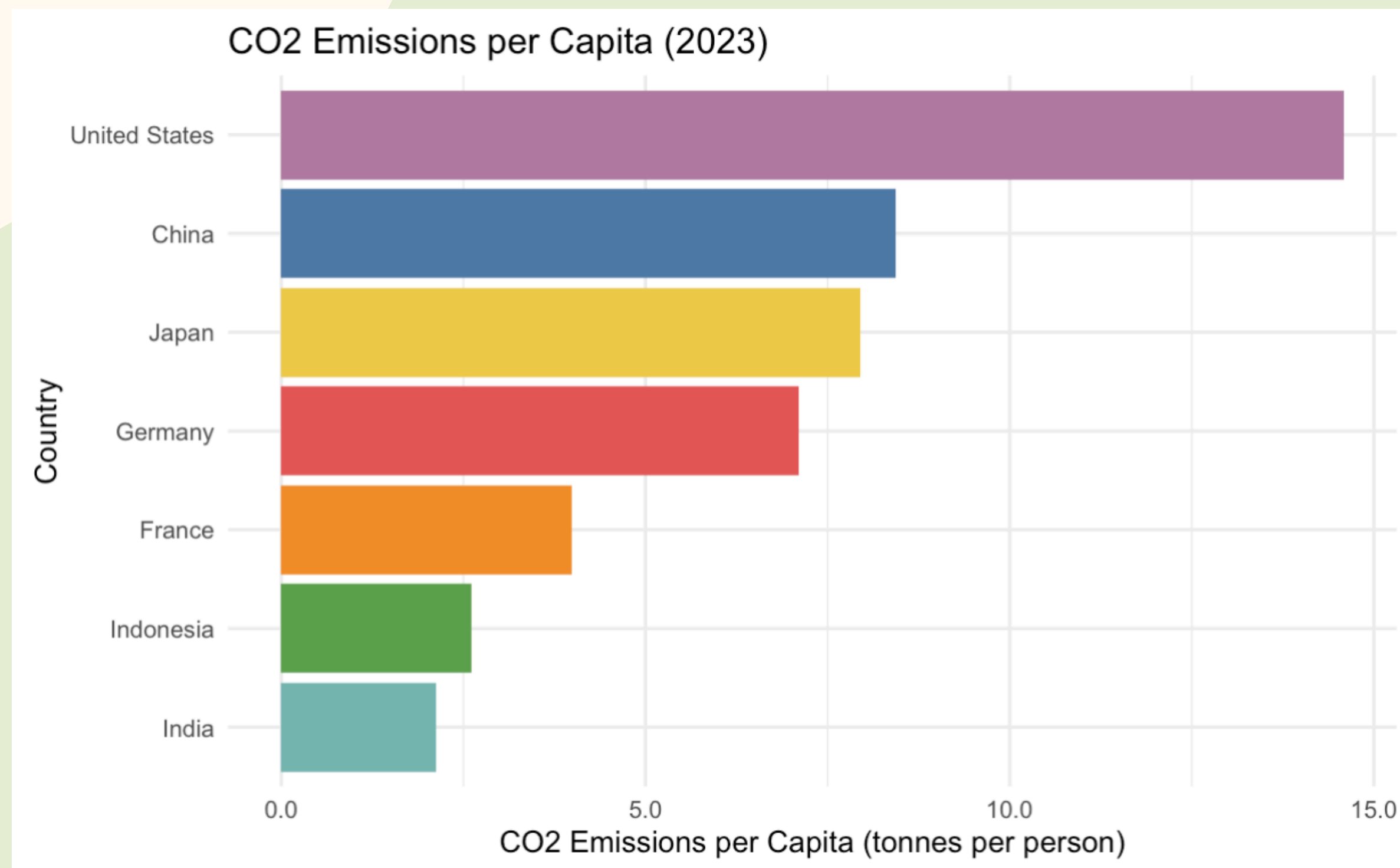


Q2: WHICH COUNTRIES HAVE EMITTED THE MOST CARBON DIOXIDE IN TOTAL?



KEY TAKEAWAY:
EUROPE AND US
HOLD HISTORICAL
RESPONSIBILITY

Q3: WHICH COUNTRIES EMIT THE MOST CARBON DIOXIDE PER CAPITA?



**KEY
TAKEAWAY:
USA + CHINA
+ JAPAN
EMIT MOST
PER
PERSON**

THE PATH FORWARD...



CONCLUSION: → CHINA, EUROPE & USA HOLD CURRENT DAY RESPONSIBILITY
→ EUROPE & USA HOLD HISTORICAL RESPONSIBILITY

DEVELOPED COUNTRIES:

USE **RESOURCES (\$)**, **SKILLED WORKFORCE** AND **TECHNOLOGY** TO DEVELOPED **LOW-COST LOW-CARBON SOLUTIONS** FOR DEVELOPING COUNTRIES

DEVELOPING COUNTRIES:

FOCUS ON **SUSTAINABLE DEVELOPMENT**, EXPAND WITH **RENEWABLE RESOURCES** AND **LOW-CARBON SOLUTIONS**





THANK YOU!
QUESTIONS?